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ORIGINAL ARTICLE

Dual asymmetric centrifugation to prepare cosmetic emulsions and creams: Physico-chemical and energy considerations

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Abstract

Objective: This study evaluates room-temperature dual asymmetric centrifugation (DAC) as an R&D scale alternative mixing technology for cosmetic oil-in-water emulsions and structured creams (200 g), compared with high-shear rotor–stator and thermo-mechanical processes, with a focus on product properties and energy requirements.

Methods: Formulations were characterized by visual assessment, conductivity and pH monitoring over 30 days. Physical stability was analysed by multiple light scattering. Flow and oscillatory measurements were performed, complemented by texture analysis. Specific energy consumption was evaluated at both scales.

Results: All systems exhibited homogeneous oil-in-water structures with no significant differences in pH evolution or physical stability. Shear-thinning behaviour was observed for all formulations, while creams showed dominant elastic behaviour ($G' > G''$), indicating structured networks independent of the mixing process. Texture properties were similar across technologies. Energy analysis revealed a scale effect: rotor–stator was more efficient for 30 g samples, while DAC became competitive at 200 g due to the absence of heating. Final product properties remained comparable across processes.

Conclusion: DAC is a robust and energy-efficient alternative for cosmetic emulsions and creams, enabling comparable product properties while offering potential energy savings at larger scales, supporting more sustainable processing strategies.

KEYWORDS
creams, dual asymmetric centrifugation, emulsions, energy efficiency, formulation processes

Résumé

Objectif: Cette étude évalue la centrifugation asymétrique double à température ambiante en tant que technologie de mélange alternative à l'échelle de la R&D

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pour les émulsions cosmétiques de type « huile dans l'eau » (30 g) et les crèmes structurées (200 g), en comparaison avec les procédés à haut cisaillement de type rotor-stator et les procédés thermomécaniques, en mettant l'accent sur les propriétés des produits et les besoins énergétiques.

Méthodes: Les formulations ont été caractérisées par une évaluation visuelle, la mesure de la conductivité et le suivi du pH sur une période de 30 jours. La stabilité physique a été analysée par diffusion multiple de la lumière. Des mesures d'écoulement et oscillatoires ont été réalisées, complétées par une analyse de texture. La consommation d'énergie spécifique a été évaluée aux deux échelles.

Résultats: Tous les systèmes ont présenté des structures de type « huile dans l'eau » homogènes, sans différences significatives quant à l'évolution du pH ou à la stabilité physique. Un comportement de fluidification sous cisaillement (rhéofluidification) a été observé pour toutes les formulations, tandis que les crèmes ont présenté un comportement élastique dominant ($G' > G''$), indiquant l'existence de réseaux structurés indépendants du procédé de mélange. Les propriétés de texture étaient similaires quelle que soit la technologie utilisée. L'analyse énergétique a révélé un effet d'échelle : le système de type rotor-stator s'est avéré plus efficace pour les échantillons de 30 g, tandis que la centrifugation asymétrique double est devenue compétitive à partir de 200 g en raison de l'absence d'échauffement. Les propriétés du produit final sont restées comparables quel que soit le procédé utilisé.

Conclusion: La centrifugation asymétrique double constitue une alternative robuste et économe en énergie pour les émulsions et les crèmes cosmétiques, permettant d'obtenir des propriétés de produit comparables tout en offrant un potentiel d'économies d'énergie à plus grande échelle, ce qui favorise des stratégies de fabrication plus durables.

INTRODUCTION

The formulation of emulsified cosmetic products, including both fluid emulsions and structured creams, relies on the combined control of formulation composition and processing conditions [1]. Cosmetic matrices require precise control of their microstructure to ensure stability, texture and the sensory properties expected by consumers [2]. The choice of mixing technology is a key parameter, influencing droplet dispersion, viscoelastic network formation and the mechanical cohesion of the final product [3, 4]. Oil-in-water emulsions are widely used in cosmetic and pharmaceutical applications, enabling the delivery of active ingredients, skin hydration and the development of light and pleasant textures [5–7]. The average droplet size and size distribution directly govern physical stability, viscosity, rheological behaviour, and ultimately sensory perception [8, 9]. In the case of creams, structuring relies on the formation of a continuous viscoelastic network, often supported by the crystallization of lipophilic structuring agents or by interactions between droplets and polymers in the aqueous phase, which confer firmness, cohesion and spreadability [10–12]. Conventionally, laboratory-scale emulsification processes rely on high-shear mixers capable of generating strong velocity gradients, leading to fine and homogeneous systems [13, 14]. However, these systems often require preheating of the phases to promote the dissolution of ingredients and reduce viscosity, thereby facilitating dispersion. As a result, mechanical and thermal inputs are typically combined [15, 16]. This thermo-mechanical approach can influence crystallization kinetics and the internal organization of the matrices, while also contributing to significant energy consumption [16, 17]. Beyond energy considerations, reducing or eliminating heating steps is particularly relevant for the incorporation of thermally sensitive cosmetic ingredients. Many active compounds, such as vitamins, antioxidants, peptides, botanical extracts, or certain fragrance components, may undergo degradation, oxidation, volatilization, or loss of activity when exposed to elevated processing temperatures. The

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development of low-temperature emulsification strategies therefore represents an important challenge for the cosmetic industry, as it may improve ingredient preservation while maintaining formulation performance and product quality [18]. In an industrial context increasingly driven by sustainability and energy efficiency requirements, the optimization of formulation processes has become a major challenge [19]. Reducing heating steps, simplifying processing protocols and ensuring adaptability across different batch sizes are now key criteria in the design of more efficient and flexible processes [20]. Dual asymmetric rotation mixers, based on a combination of planetary motion and centrifugation, represent a promising alternative technology. This approach generates intensive mixing through the simultaneous rotation and revolution of the container, without the need for immersed shear tools [21, 22]. It enables processing at ambient temperature, thereby limiting thermal input and is suitable for a wide range of batch sizes (from 10 g to 15 kg). In addition, reduced air incorporation contributes to improved system stability and to the preservation of heat- or oxidation-sensitive ingredients [23]. Despite growing interest in these technologies, no studies have systematically evaluated their ability to structure cosmetic matrices of increasing complexity, ranging from fluid emulsions to structured creams, in comparison with conventional processes. In particular, the influence of mixing technology on microstructure, viscoelastic and textural properties, stability and energy consumption remains insufficiently documented. In the present study, the term 'emulsion' refers to a fluid oil-in-water dispersed system with limited structural organization, while 'structured cream' designates an oil-in-water emulsion exhibiting a developed viscoelastic network and enhanced textural properties. This study proposes an integrated evaluation of dual asymmetric centrifugation as a mixing technology for cosmetic formulation. It aims to compare the performance of conventional processes and DAC across two levels of complexity: the preparation of fluid emulsions and the formulation of structured creams. In parallel, a comparative analysis of energy consumption was carried out, considering both the mechanical energy delivered by the mixing devices and, where applicable, the thermal energy required for phase heating.

MATERIALS AND METHODS

Chemicals

Polysorbate 80, sorbitan monooleate and Solagum AX® (a blend of Acacia senegal gum and xanthan gum) were kindly supplied by SEPPIC (Paris, France). Sunflower oil was purchased from IES Labo (Oraison, France). Benzyl alcohol, dehydroacetic acid, betaine, glycerol and citric acid were obtained from Fisher Scientific (Illkirch, France). Eumulgin SG® (sodium stearoyl glutamate) was provided by AMI Ingrédients (Tauxigny, France). Miglyol® 812N (caprylic/capric triglycerides) was supplied by IOI Oleo GmbH (Hamburg, Germany). The water used for emulsion preparation was purified using a Milli-Q system (Millipore Corporation, Bedford, MA, USA).

Formulation process and composition

Emulsion

Oil-in-water (O/W) emulsions were formulated using the same composition for both mixing technologies (Table 1), expressed as weight percentages (% w/w). The aqueous phase accounted for 76.25% w/w of the formulation and consisted of purified water, polysorbate 80 (1.25% w/w) and a gelling agent blend of Acacia senegal gum and xanthan gum (Solagum AX, 0.50% w/w). A preservative system based on benzyl alcohol and dehydroacetic acid (1.00% w/w) was incorporated after emulsification during cooling at approximately 40°C in the thermo-mechanical process. The oil phase represented 23.75% w/w of the formulation and was composed of sunflower oil (20.0% w/w) and sorbitan oleate (3.75% w/w). Each experimental condition was performed in six independent replicates ($n = 6$). For preparation using high-shear mixing (Ultra-Turrax® T25), 30 g batches were obtained after separately heating the aqueous and oil phases to 70°C. The aqueous phase was gradually added to the oil phase under stirring, followed by homogenization at 20 000 rpm for 2 min. When the gelling agent was incorporated into the aqueous phase prior to emulsification, an additional mixing step of 1 min

TABLE 1 Quantitative composition of the emulsion.

Ingredients	Function	%wt.
Sorbitan monooleate (%wt)	Emulsifier	3.75
Sunflower oil (%wt)	Emollient	20
Polysorbate 80 (%wt)	Emulsifier	1.25
Acacia senegal gum/xanthan gum mix (%wt)	Thickener	0.50
Dehydroacetic acid/benzyl alcohol mix (%wt)	Antimicrobial agent	1
Water (% wt)	Solvent	q.s 100

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was applied to ensure system homogeneity. After homogenization, emulsions were allowed to cool at ambient temperature before characterization. For preparation by dual asymmetric centrifugation (SpeedMixer® DAC 150), emulsions were formulated at ambient temperature without external thermal input. Batches of 30 g corresponding to the minimum mass relative to the conventional mixer used were prepared following pre-dispersion of Solagum AX in the aqueous phase at 2500 rpm for 5 min. The oil phase was then directly added to the same container, and the final mixing step was performed at 3500 rpm for 5 min. Emulsions were characterized in terms of pH, conductivity, droplet size distribution and microstructural observation by optical microscopy. Microstructural observations were performed using an Olympus iX51 optical microscope (Olympus, Japan) coupled with an Olympus DP72 digital camera. Samples were observed without prior dilution at a magnification of ×400.

Cream

Creams were formulated using a single composition (Table 2), expressed as weight percentages (% w/w), and prepared in 200 g batches. Each process was carried out in six independent replicates ($n = 6$). The aqueous phase consisted of water, betaine (4% w/w), glycerol (3% w/w), and a pH adjuster (citric acid, 1% w/w). The preservative system (1% w/w) was incorporated after emulsification during cooling at approximately 40°C in the thermo-mechanical process. The oil phase was composed of caprylic/capric triglycerides (12% w/w) and sodium stearoyl glutamate (3% w/w). A gelling system (Acacia senegal gum/xanthan gum, 1% w/w) was used as a structuring agent for the continuous phase. For the thermo-mechanical process using a deflocculating mechanical mixer (TurboTest® VMI), the aqueous and oil phases were heated separately to 70°C. Emulsification was carried out by gradually incorporating the oil phase into the aqueous phase under stirring at 2500 rpm for 10 min, followed by a stabilization step at 1500 rpm for 10 min. The gelling agent was then added under continuous stirring at 1500 rpm for an additional 10 min to allow progressive structuring of the system during cooling at ambient temperature. For the dual asymmetric centrifugation process (SpeedMixer® SMART DAC), preparation was performed at ambient temperature without prior heating of the phases. The gelling agent was first pre-mixed with glycerol at 800 rpm for 2 min to promote dispersion. The remaining components of the aqueous phase were then added, and mixing was continued for 2 min 30 s at 2000 rpm. The oil phase was subsequently added directly to the same container, followed by a final emulsification step for 5 min at 2000 rpm. The resulting creams were characterized in terms of pH, conductivity, droplet size distribution and microstructural observation. Microstructural observations were performed using an Olympus iX51 optical microscope (Olympus, Japan) coupled with an Olympus DP72 digital camera. Samples were observed without prior dilution at a magnification of ×400.

TABLE 2 Quantitative composition of the cream.

Ingredients	Function	%wt.
Sodium stearoyl glutamate (%wt)	Emulsifier	3
Caprylic/capric triglycerides (%wt)	Emollient	12
Acacia senegal gum/xanthan gum mix (%wt)	Thickener	1
Glycerin (%wt)	Humectant	3
Betaine (%wt)	Humectant	4
Citric acid	pH agent	1
Dehydroacetic acid/ benzyl alcohol mix (%wt)	Antimicrobial agent	1
Water (% wt)	Solvent	q.s 100

Static multiple light scattering-based stability analysis

Formulation stability was evaluated using static multiple light scattering with a Turbiscan® instrument (Microtrac, Toulouse, France). Analyses were performed in 4 mL glass vials, 24 h after preparation (Day 1) and after 30 days of storage (Day 30). The Turbiscan Stability Index (TSI) was calculated using the associated software to estimate the overall stability of the systems. An estimation of the mean droplet size was also derived from the backscattering signal.

Rheology

Rheological properties were determined using a Kinexus Pro+ rheometer (Netzsch, Germany). Emulsion measurements were performed using a sand-blasted Couette (cylindrical) geometry to minimize wall slip. After loading, samples were allowed to rest for at least 5 min at 25°C to ensure structural relaxation and thermal equilibration prior to analysis. Apparent viscosity was measured using an increasing shear rate ramp from 0.1 to 1000 s^{-1} over a total duration of 30 min, with 10 measurement points per decade. Creams were analysed using sand-blasted cone-and-plate geometry (4° angle), suitable for structured

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semi-solid systems. Flow measurements were conducted under the same conditions as for emulsions (shear rate ramp from 0.1 to 1000 s^{-1}, 10 points per decade, 25°C). Oscillatory tests were performed to characterize the viscoelastic behaviour of the creams. An amplitude sweep was conducted at a constant frequency of 1 Hz, with strain ranging from 0.01% to 100% (10 points per decade), allowing determination of the linear viscoelastic region (LVER). A frequency sweep was then carried out within the LVER, at a fixed strain of 0.5%, over a frequency range of 0.1 to 50 Hz (10 points per decade). Storage (G′) and loss (G″) moduli were recorded to assess structural organization and elastic dominance.

Textural analysis

The textural properties of the creams were evaluated using an EZ-SX texture analyser (Shimadzu, Japan) at ambient temperature (25°C). Compression tests were performed using a cylindrical acrylic probe (12.7 mm diameter, 1/2 inch), commonly used for standardized gel testing. Samples were placed in cylindrical containers (35 mm diameter) and filled to a standardized height of 15 mm prior to analysis. The probe penetrated the sample from 0 to 5 mm at a constant speed of 10 $mm \cdot min^{-1}$. A trigger force of 0.005 N was applied to ensure reproducible contact between the probe and the sample surface. From the resulting force–distance curves, the following parameters were determined: hardness (maximum force during compression), springiness, cohesiveness and adhesiveness, the latter corresponding to the negative area measured during probe withdrawal. Data were processed using Trapezium-V software (Shimadzu).

Energy consumption analysis

The energy requirements of the different mixing processes were estimated using a global approach combining the mechanical energy supplied by the mixing devices and, where applicable, the thermal energy required for phase heating. This approach is commonly used to compare the energy efficiency of processes in chemical engineering and formulation science [24].

Mechanical energy consumption was estimated from the nominal electrical power of the equipment and the effective operating time, according to:

$$E_{\text{mechanical}} = P \times t$$

where P is the electrical power (W), and t is the processing time (s) required to prepare one batch. This provides a comparative estimate under controlled experimental conditions.

For processes involving heating (30 g emulsions and 200 g creams prepared by the thermo-mechanical method), thermal energy was estimated using:

$$E_{thermal} = m \times C_{\text{p}} \times \Delta T$$

where m is the mass of the heated phase (kg), C_{p} is its specific heat capacity ($J kg^{-1} K^{-1}$), and ΔT is the temperature increase from 25°C (298.15 K) to 70°C (343.15 K). The specific heat capacity of water ($C_{\text{p}} \approx 4180 J kg^{-1} K^{-1}$) was used as an approximation for the aqueous phase, representing approximately 75% of the total formulation mass. This assumption allows consistent comparison between processes. Heat losses to the environment were not considered, as the objective was to obtain a comparative rather than absolute energy assessment [25]. For dual asymmetric centrifugation processes, formulations were prepared at ambient temperature; therefore, only mechanical energy was considered. Total energy consumption was calculated as:

$$E_{\text{total}} = E_{\text{mechanical}} + E_{\text{thermal}}$$

The obtained values were normalized by the mass of the formulation ($J g^{-1}$) to enable direct comparison between mixing technologies and batch scales (30 g vs. 200 g).

Statistical analysis

Results are expressed as mean ± standard deviation (SD). Statistical comparisons were performed using non-parametric Kruskal–Wallis and Mann–Whitney tests, as appropriate, with GraphPad Prism version 8.0. Rheological parameters were determined individually for each replicate using the rSpace software provided with the rheometer and subsequently averaged using R software.

RESULTS AND DISCUSSION

Formulation

Regardless of the mixing technology used (high-shear rotor–stator mixer (Ultra-Turrax T25), conventional mechanical stirrer (TurboTest), or dual asymmetric centrifugation (DAC)), all formulations exhibited, immediately after preparation, a homogeneous white appearance, with no visible signs of phase separation. The white colouration observed for all samples is typical of oil-in-water (O/W) emulsions with finely dispersed oil droplets in a continuous aqueous phase, where light scattering and backscattering at interfaces contribute to this macroscopic appearance [26]. Conductivity measurements (around or above 50 $\mu S cm^{-1}$) confirmed this interpretation. For all formulations,

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whether prepared with Ultra-Turrax T25, TurboTest, or DAC, strictly positive conductivity values were measured at Day 1 (D1) and maintained at Day 30 (D30). The presence of measurable conductivity indicates the continuity of the aqueous phase, thereby confirming the oil-in-water nature of the systems. No significant differences were observed between the different mixing technologies for a given formulation. Electrical conductivity (around or above 50 $\mu S\,cm^{-1}$) is a robust criterion for identifying the continuous phase of emulsions, as measurable values are associated with a conductive aqueous continuous phase. These results confirm that the mixing technology does not influence the type of emulsion obtained here (oil-in-water), provided that the formulation is identical [27, 28]. The pH values measured 24 h after preparation (D1) and after 30 days of storage (D30) are presented in Table 3. For 30 g emulsions prepared using Ultra-Turrax T25 and DAC, initial pH values ranged from 6.8 to 6.9. For 200 g creams obtained using TurboTest and SMART DAC, initial values were slightly lower, around 5.9 to 6.0. After 30 days of storage, a moderate decrease in pH was observed for all samples: emulsions showed values between 5.6 and 5.8, while creams ranged from 5.2 to 5.4. This evolution was comparable across all mixing technologies, and no statistically significant differences were detected (Kruskal–Wallis test, $p > 0.05$). The slight decrease in pH during storage may be attributed to the presence of the preservative system containing a weak organic acid, which may induce a gradual acidification of the aqueous phase [29]. Similar behaviours have been reported for emulsions preserved with organic acid-based systems, without significant impact on physical stability [29–31]. Overall, these results indicate that, despite differences in processing conditions, particularly the use of a heating step for rotor–stator and mechanical stirring processes, in contrast to DAC performed at ambient temperature, global physicochemical parameters such as emulsion type, conductivity and pH evolution remain comparable. Any potential differences between mixing technologies are therefore not expressed at the macroscopic or initial chemical level, but must be investigated at a finer scale, particularly in terms of microstructure, colloidal stability and rheological structuring.

Static multiple light scattering-based analysis

Physical stability of the formulations was evaluated by static multiple light scattering (SMLS) using a Turbiscan®. Backscattering profiles were recorded at 24 h (D1) and after 30 days of storage (D30), and overall stability was quantified using the Turbiscan Stability Index (TSI). For emulsions (30 g), backscattering variations remained low across the sample height, ranging from 0.73% to 1.60% depending on the mixing technology (Table 4). These low variations indicate the absence of macroscopic destabilization phenomena such as creaming, sedimentation, or coalescence over 30 days [32]. Initial TSI values at D1 were identical for Ultra-Turrax and DAC 150 (0.2 ± 0.1), indicating high and comparable initial stability between technologies [2]. Statistical analysis showed no significant difference between devices (Mann–Whitney test, $p = 0.33$). The mean droplet size estimated from the backscattering signal showed a slight increase during storage (4.3–5.1 µm at D1 versus 5.2–5.7 µm at D30). This moderate evolution suggests very limited destabilization kinetics, likely associated with minor coalescence or interfacial rearrangement phenomena, without impact on macroscopic stability. These

TABLE 3 pH and conductivity of the formulation at D1 and D30. Data represent the mean ± SD.

	Conditions	pH D1	pH D30	Conductivity D1 $\mu S\,cm^{-1}$
Emulsion (30 g)	Ultra-Turrax T25	6.8 ± 0.2	5.6 ± 0.3	292.3 ± 19.2
	DAC 150	6.9 ± 0.1	5.8 ± 0.4	311.8 ± 09.2
Cream (200 g)	TurboTest	6.0 ± 0.4	5.4 ± 0.3	310.7 ± 31.9
	SMART DAC	5.9 ± 0.4	5.2 ± 0.4	264.7 ± 06.7

TABLE 4 Backscattering variation, Turbiscan Index Stability (TSI) and droplet diameter of formulations. Data represent the mean ± SD.

		Backscattering variation (%)		TSI	Droplets diameter (µm)	
	Conditions	D1	D30	D1	D1	D30
Emulsion (30 g)	Ultra-Turrax	0.73 ± 0.22	0.87 ± 0.24	0.2 ± 0.1	4.3 ± 0.4	5.2 ± 0.3
	DAC 150	1.23 ± 0.31	1.60 ± 0.44	0.2 ± 0.1	5.1 ± 0.2	5.7 ± 0.4
Cream (200 g)	TurboTest	0.57 ± 0.20	0.52 ± 0.19	0.1 ± 0.1	4.0 ± 0.1	4.9 ± 0.6
	SMART DAC	0.55 ± 0.09	0.48 ± 0.09	0.1 ± 0.1	6.2 ± 0.2	6.4 ± 0.4

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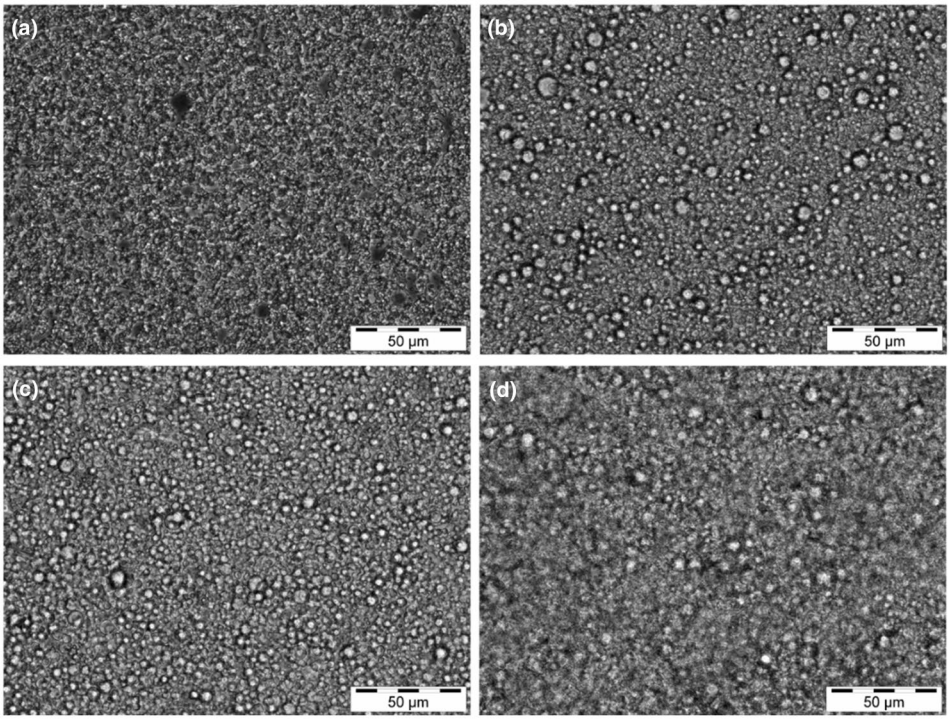


FIGURE 1 Optical microscopy at D1 of emulsions obtained with Ultra-Turrax (a), DAC 150 (b) and creams obtained with TurboTest (c), SMART DAC (d).

results indicate that DAC processing generates emulsions with microstructure and kinetic stability comparable to those obtained by conventional high-shear mixing. For creams (200 g), backscattering variations were even lower, ranging from 0.48% to 0.57%, reflecting enhanced stability due to higher viscosity and structuring of the continuous phase resulting from the presence of a significant concentration of thickeners in the aqueous phase. Initial TSI values at D1 were 0.1 ± 0.1 for both technologies (TurboTest and DAC), with no significant difference (Mann–Whitney test, $p = 0.67$). Mean droplet diameters showed a slight increase between D1 (4.0–6.2 µm) and D30 (4.9–6.4 µm), but within limited proportions. The more structured matrix of creams appears to restrict droplet mobility and slow down gravitational separation mechanisms. Although SMART DAC samples exhibited slightly higher droplet diameters, no statistically significant difference was observed, confirming that mixing technology does not significantly affect the final microstructure of the system. Optical microscopy observations (Figure 1) corroborate the SMLS results, showing homogeneous distributions and spherical droplets for all formulations, regardless of the mixing process. Overall, whether for fluid emulsions or more structured creams, DAC processing at ambient temperature and without external heating enables the production of physically stable systems over 30 days, comparable to those obtained with conventional high-shear mixers. These observations confirm that mixing technology does not alter

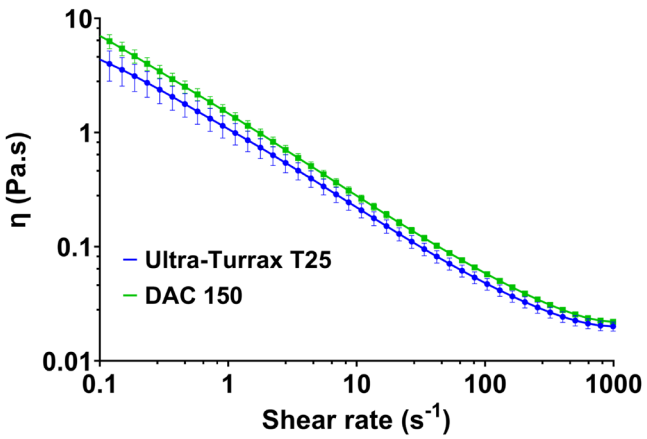


FIGURE 2 Flow properties of emulsions obtained with Ultra-Turrax T25 (blue) and DAC 150 (green).

kinetic stability and are consistent with the macroscopic and pH results described in the previous section.

Rheology

Viscosity measurements of emulsions and creams allowed evaluation of flow and oscillatory rheological behaviour for the different mixing devices. For 30 g oil-in-water emulsions, viscosity curves as a function of shear rate (Figure 2) showed shear-thinning behaviour for both technologies

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TABLE 5 Value of viscosity at D1 (in Pa.s) obtained at different shear rates (in s^{-1}) during the flow test of emulsions studied.

Conditions	η (0.1)	η (1)	η (10)	η (100)	η (1000)
Ultra-Turrax T25	4.48 ± 1.38	1.15 ± 0.25	0.21 ± 0.03	0.05 ± 0.01	0.02 ± 0.00
DAC 150	7.27 ± 1.05	1.58 ± 0.18	0.26 ± 0.02	0.06 ± 0.01	0.02 ± 0.00

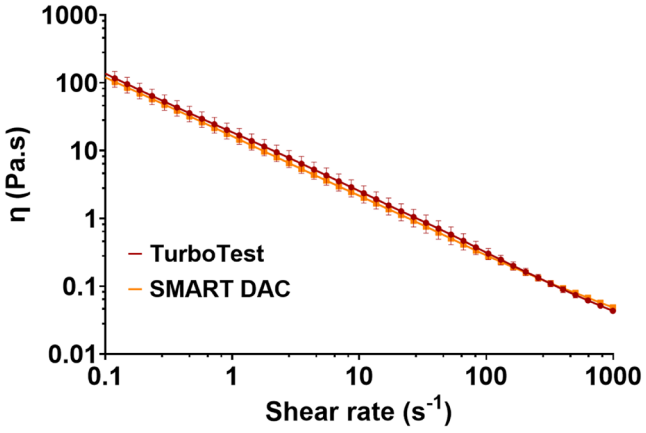


FIGURE 3 Flow properties of creams obtained with TurboTest (red) and Smart DAC (orange).

studied. Apparent viscosity values measured at different shear rates are presented in Table 5. Differences between Ultra-Turrax and DAC 150 were not statistically significant (Mann–Whitney test, $p = 0.42$), and both technologies produced fluid and homogeneous emulsions with comparable flow properties [33, 34].

For 200 g creams, rheological flow behaviour (Figure 3) also showed shear-thinning behaviour, with a progressive decrease in apparent viscosity as shear rate increased, typical of structured systems. Viscosity values at different shear rates are reported in Table 6. Differences between TurboTest and SMART DAC remained modest and were not statistically significant (Mann–Whitney test, $p = 0.88$), indicating that both technologies produce creams with comparable viscosity despite different processing conditions (heating and mixing intensity).

Oscillatory measurements performed on creams allowed characterization of their viscoelastic structure. All creams exhibited a well-defined linear viscoelastic region (LVR) and a predominantly elastic behaviour. The storage modulus (G′) was higher than the loss modulus (G″) for both technologies, with a tan δ (G″/G′) lower than 1, confirming elastic dominance and mechanical cohesion of the creams. The data are summarized in Table 7. The G′ value within the LVR was almost twice as high for TurboTest compared to SMART DAC, suggesting slightly stronger structuring in thermally processed creams. The yield stress observed for TurboTest was 45 Pa, indicating the presence of a stress threshold required to initiate flow, whereas no yield stress was detected for SMART DAC under the tested conditions. This absence is likely related to a different deployment of the gelling agent due to differences in shear conditions during mixing.

These results indicate that both devices produce structured and cohesive creams with comparable viscoelastic behaviour. The heating step in TurboTest slightly enhances network rigidity (higher G′), but differences remain moderate and do not compromise the overall similarity of rheological properties.

Textural analysis

Textural analysis of creams was performed to complement information on structuring and viscoelastic behaviour. Measured properties included hardness, cohesiveness and adhesiveness (Table 8). Cream hardness was similar for both mixing technologies, with very low values, reflecting the easily deformable nature of the formulated creams. Cohesiveness showed a noticeable difference: creams prepared using SMART DAC exhibited higher cohesiveness than those prepared with TurboTest, suggesting a slightly more homogeneous or deformation-resistant internal network. Adhesiveness was low and comparable between both processes, indicating that the force required to detach the probe from the sample was minimal, consistent with the relatively fluid nature of the creams studied. These results confirm that, despite differences in processing conditions (heating and mixing time for TurboTest versus ambient-temperature processing for SMART DAC), creams exhibit globally comparable textural properties suitable for cosmetic applications [35, 36]. The slightly higher cohesiveness observed for SMART DAC may reflect the efficiency of rotational–centrifugal mixing in uniformly structuring the matrix without thermal input, consistent with oscillatory rheology results indicating a predominantly elastic viscoelastic behaviour.

Energy consumption analysis

Energy requirements associated with the different mixing technologies were evaluated to complete the performance analysis and identify potential trade-offs between product quality and process efficiency. Total energy consumption was estimated by considering both mechanical energy from the mixing devices and, where applicable,

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TABLE 6 Value of viscosity at D1 (in Pa.s) obtained at different shear rates (in s^{-1}) during the flow test of creams studied.

Conditions	η (0.1)	η (1)	η (10)	η (100)	η (1000)
TurboTest	144.00 ± 37.78	20.20 ± 5.29	2.35 ± 0.66	0.30 ± 0.06	0.04 ± 0.00
SMART DAC	126.45 ± 13.71	17.85 ± 1.83	2.01 ± 0.15	0.28 ± 0.02	0.05 ± 0.00

TABLE 7 Rheological data (means ± SD) obtained from the oscillatory tests of the creams studied.

Conditions	Linear viscoelastic region (LVR) (Pa)	Flow threshold (Pa)	$\tan\delta = G''/G'$
TurboTest	2.89 ± 0.26	45.21 ± 7.34	0.35 ± 0.02
SMART DAC	4.58 ± 0.32	N/A	0.33 ± 0.04

TABLE 8 Textural data (means ± SD) obtained for creams.

Conditions	Hardness (*N*)	Springness	Cohesiveness	Adhesiveness
TurboTest	0.02 ± 0.00	N/A	0.50 ± 0.12	0.00 ± 0.02
SMART DAC	0.02 ± 0.01	N/A	1.00 ± 0.10	0.00 ± 0.04

TABLE 9 Summary of process parameters and estimated energy consumption for the different formulation technologies.

Conditions	Batch size (g)	Mixing time (min)	Heating step	Total energy (kJ)	Specific energy ($kJ\,g^{-1}$)
Ultra-Turrax T25	30	3	Yes (25°C→70°C)	148	4.9
DAC 150	30	10	No	570	19.0
TurboTest	200	30	Yes (25°C→70°C)	900	4.5
SMART DAC	200	9.5	No	675	3.4

thermal energy required for phase heating [16]. For DAC devices, manufacturer-reported maximum powers (950 W for DAC 150 and 1500 W for SMART DAC) were used, combined with total mixing time. This conservative approach is consistent with methods commonly used for comparing laboratory-scale mixing processes [37]. For 30 g emulsions, the Ultra-Turrax T25 high-shear process required prior heating of phases and 3 min mixing at 20 000 rpm. Mechanical energy was estimated as 800 W × 3 min, and thermal energy was calculated based on heating the aqueous phase from 25°C to 70°C. The DAC 150, used for equivalent batch sizes at ambient temperature, required 10 min of mixing at rotational speeds between 2500 and 3500 rpm (nominal power 950 W). Total and specific energy values are reported in Table 9. Although DAC 150 exhibited higher specific energy consumption at small scale, both technologies produced emulsions with comparable stability, droplet size and rheological properties. For 200 g creams, a comparison was made between TurboTest VMI (thermo-mechanical process) and SMART DAC. TurboTest involved heating to 70°C, followed by emulsification at 2500 rpm for 10 min, stabilization at 1500 rpm for 10 min and gelling agent incorporation under stirring for an additional 10 min. With a power consumption of 400 W, both mechanical and thermal energies were estimated accordingly. SMART DAC enabled preparation at ambient temperature without heating, using three successive mixing steps (2 min at 800 rpm, 2 min 30 s at 2000 rpm and 5 min at 2000 rpm), with a nominal power of 1500 W. The calculated specific energy reflects improved mixing efficiency for larger batch sizes while avoiding thermal input. These results highlight a scale effect: at low batch mass, specific energy consumption is higher, particularly for high-shear or long-duration processes, whereas increasing batch size reduces specific energy consumption while maintaining equivalent product quality. Differences in energy consumption do not affect final emulsion or cream properties, as confirmed by stability (TSI, backscattering), droplet size, pH, rheology and texture. Finally, these results underline the interest of dual asymmetric centrifugation technologies for larger-scale formulations, enabling a significant reduction in specific energy consumption while eliminating heating steps. This approach aligns with current eco-design strategies in cosmetics and pharmaceuticals, balancing product performance and process energy efficiency.

General discussion

Overall, the results demonstrate that the different mixing technologies studied, high-shear rotor–stator,

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thermo-mechanical agitation and dual asymmetric centrifugation (DAC), produce oil-in-water emulsions and creams with globally comparable physicochemical properties when systems are considered pairwise. At the macroscopic level, emulsions prepared by rotor–stator and DAC show similar characteristics in terms of appearance, conductivity and pH evolution during storage. Likewise, creams obtained by thermo-mechanical agitation and DAC do not exhibit significant differences in these parameters. These observations are consistent with literature reporting that the continuous phase nature and overall stability of O/W emulsions depend primarily on formulation once sufficient dispersion is achieved [38]. Differences between technologies emerge mainly at microstructural and mechanical levels. Rheological tests show that all formulations exhibit shear-thinning behaviour characteristic of concentrated dispersed systems [35, 39]. Oscillatory measurements confirm the formation of structured viscoelastic networks in both cream formulations prepared using conventional high-shear mixing and dual asymmetric centrifugation, although thermo-mechanical systems show a slightly more extended linear viscoelastic region, suggesting a more rigid structure. This behaviour can be attributed to the combined effect of shear and thermal cycling, promoting partial crystallization of the lipid phase and strengthening of the network. These differences are linked to the specific mixing mechanisms of each technology. Rotor–stator systems rely on localized high-shear-generating droplet fragmentation through high velocity gradients [40], whereas dual asymmetric centrifugation generates a complex force field combining centrifugal acceleration and multiple relative motions, leading to a more homogeneous distribution of mechanical stresses throughout the sample [22]. Thermo-mechanical processing combines agitation and heat input, influencing both dispersion and phase transitions. These mechanistic differences explain how similar macroscopic properties can arise from slightly different microstructures. Textural measurements confirm these trends. For each pair of processes, hardness and adhesiveness remain comparable, while moderate variations in cohesiveness are observed. These differences remain within typical ranges reported for cosmetic emulsions of similar composition [41]. Energy analysis provides complementary insight. At small scale (30 g), rotor–stator mixing is more efficient than DAC due to shorter processing time, despite the need for heating. In contrast, at a larger scale (200 g), SMART DAC becomes more efficient than the thermo-mechanical process, mainly due to the elimination of the heating step. This behaviour is consistent with studies showing that non-thermal processes gain energetic advantage as scale increases, particularly when heat transfer limitations become significant [42]. Overall, these results demonstrate that process performance is not solely governed by shear intensity but also by the nature and distribution of mechanical stresses and thermal management. In this context, DAC represents a relevant alternative to conventional methods, particularly at intermediate to large batch sizes where its energy efficiency becomes competitive or even advantageous. From an eco-design perspective, these findings highlight the importance of simultaneously considering product quality parameters (stability, rheology, texture) and energy indicators. The ability to formulate at ambient temperature while maintaining comparable structural properties to heated processes constitutes a significant advantage for reducing the energy footprint of cosmetic formulation steps. In conclusion, dual asymmetric centrifugation appears as a robust and energy-relevant technology for cosmetic emulsions and creams, achieving physicochemical performance comparable to conventional processes while offering opportunities for process energy optimization.

CONCLUSION

This study compared the technological and energy efficiency of different mixing approaches for the formulation of emulsions (30 g) and creams (200 g). The results show that, regardless of the process used, stable, homogeneous oil-in-water systems with comparable physicochemical properties can be obtained. Stability, rheological and textural analyses indicate that differences induced by mixing technologies remain limited and do not significantly affect final formulation performance. Rheological tests confirm shear-thinning behaviour for all systems and a dominant elastic modulus in creams, reflecting the formation of comparable structured networks. Observed variations in rigidity or cohesiveness remain moderate and compatible with standard cosmetic applications. Energy analysis, however, highlights a clear scale effect. While conventional mixing appears more efficient at very small scales, dual asymmetric centrifugation becomes competitive or even advantageous at larger batch sizes, particularly due to the elimination of heating steps. From a process optimization and energy reduction perspective, dual asymmetric centrifugation therefore represents a robust alternative to conventional methods, enabling the balance between product performance and energy efficiency in cosmetic emulsion formulation.

AUTHOR CONTRIBUTIONS

The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

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CONFLICT OF INTEREST STATEMENT

None.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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